

International Islamic University Chittagong (IIUC)
 Department of Computer Science and Engineering (CSE)
 Mid Term Examination

Program: B. Sc. in CSE
 Course Code: MATH-1107
 Time: 1:30 hours



Semester: Spring 2026
 Course Title: Mathematics-I
 Total Marks: 30

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
 (ii) Please answer the several parts of a question sequentially.
 (iii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

CLO1:	Compute the functions, limit and continuity of a function, derivatives, integrals and extrema of single-variable and/or multivariable functions.
CLO2:	Understand the techniques of differentiation and integration.

Bloom's Taxonomy Domain Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

- | | Marks | CLO | DL |
|--|-------|------|----|
| 1. a) Find the domain and range of the function, $f(x) = \sqrt{\frac{x+2}{x-2}}$ | 3 | CLO1 | R |
| b) Test the differentiability of the following function at $x = 0$
$y = f(x) = \begin{cases} 3 + 2x, & -\frac{3}{2} < x < 0 \\ 3 - 2x, & 0 \leq x < \frac{3}{2} \\ -3 - 2x, & x \geq \frac{3}{2} \end{cases}$ | 4 | CLO1 | U |
| c) Test the continuity of the following function at $x = 0$
$y = f(x) = \begin{cases} \frac{1 - \cos x}{x^2}; & x \neq 0 \\ 1; & x = 0 \end{cases}$ | 3 | CLO1 | U |
| Or)
d) Evaluate the limit using L'Hospitals rule, $\lim_{x \rightarrow \infty} 2^x \sin\left(\frac{a}{2x}\right)$ | | | |

2. a) Find the differential co-efficient of $\log_a x$ by the geometrical method. 3 CLO2 U
- b) Differentiate $\sec^{-1} \frac{1}{2x^2-1}$ with respect to $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ 4 CLO2 U
- Or)
- c) Find the differential coefficient of the following functions with respect to x
- (i) $y = (\sin^{-1} x)^{\ln x}$ (ii) $\sqrt{\frac{x}{y}} + \sqrt{\frac{y}{x}} = 1$
- d) If $y = \cos^2 x + \frac{1}{\cos^2 x} + \frac{1}{\cos^2 x} + \frac{1}{\cos^2 x} + \dots \dots \dots \infty$ then find $\frac{dy}{dx}$ 3 CLO2 U
3. a) If $x = \cos \sqrt{y}$ then using the Leibnitz theorem show that,
 $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$ 4 CLO2 U
- b) Verify the Roll's theorem for the function $f(x) = e^x \sin x$ in the interval $(0, \pi)$. 3 CLO2 U
- Or)
- c) Verify Mean Value theorem for the function,
 $f(x) = x(x-1)(x-2)$ in the interval $(0, 0.5)$.
- d) Expand $\ln x$ in ascending power of $(x-2)$ by Taylor's Series. 3 CLO2 U

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Semester: Autumn 2025
 Course Title: Mathematics-I
 Total Marks: 30

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Bloom's Taxonomy Domain Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

	Marks	CLO	DL
1. a) Define Domain and Range of a function.	1	CLO1	R
b) Find the domain and range of the function, $f(x) = \sqrt{x^2 - 9}$	2	CLO1	U
c) Test the continuity of the following function at $x = 0$ $y = f(x) = \begin{cases} \frac{1 - \cos x}{x^2}; & x \neq 0 \\ 1; & x = 0 \end{cases}$	4	CLO1	U
Or. Test the differentiability of the following function at $x = 1$ $y = f(x) = \begin{cases} 5x - 4; & 0 < x \leq 1 \\ 4x^2 - 3x; & 1 < x < 2 \end{cases}$			
d) Evaluate the limit using L'Hospitals rules $\lim_{x \rightarrow \infty} (e^x + 1)^{\frac{-2}{x}}$	3	CLO1	U

2. a) Find the differential co-efficient of $\ln x$ by the geometrical method. 3 CLO2 U

b) Differentiate $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ with respect to $\sec^{-1} \frac{1}{2x^2-1}$ 4 CLO2 U

c) If $y = (\cos x)^{(\cos x)^{(\cos x)^{\dots \infty}}}$ then find $\frac{dy}{dx}$ 3 CLO2 U

3. a) If $x = \sin\left(\frac{1}{m} \ln y\right)$ then using the Leibnitz theorem show that, $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0$ 4 CLO2 U

b) Verify the Roll's theorem for the function $f(x) = x^2 - 3x + 2$ in the interval (1,2) 3 CLO2 U

Or, Obtain the Maclaurin's series generated by the function $f(x) = \ln(1+x)$

c) Verify Mean Value theorem for the function $f(x) = \frac{x^3}{4} + 1$ over the interval [0, 2]. 3 CLO2 U

Or. Expand $\frac{1}{x}$ in ascending power of $(x-2)$ by Taylor's Series.

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Course Title: Mathematics-I
Total Marks: 30

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
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CLO1:	Compute the functions, limit and continuity of a function, derivatives, integrals and extrema of single-variable and/or multivariable functions.
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Bloom's Taxonomy Domain Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

		Marks	CLOs	DLs
1.	a) Find the Maclaurin Series for the function $f(x) = \cos x$	2	CLO1	U
	b) Find the domain & range of the function, $f(x) = \sqrt{\frac{1-x}{x}}$	2	CLO1	U
	c) Test the differentiability of the following function at $x = 2$ $f(x) = 3x + 2; \quad x > 2$ $\quad \quad \quad = 12 - x^2; \quad x \leq 2$	6	CLO1	App
	Or Test the differentiability of the following function at $x = 2$, where $f(x) = \begin{cases} x & \text{when } 0 < x < 1 \\ 2 - x & \text{when } 1 \leq x \leq 2 \\ x - \frac{1}{2}x^2 & \text{when } x > 2 \end{cases}$	6	CLO1	App
2.	a) Find the differential co-efficient of e^x by the first principle method.	3	CLO2	U
	b) If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots - \infty}}}$; then Show that $y_1 = \frac{\cos x}{2y - 1}$	3	CLO2	U

- c) Evaluate the limit using L'Hospitals rules for $\lim_{x \rightarrow 0} \left(\frac{1}{x}\right)^{\tan x}$ 4
Or

Differentiate $\tan^{-1} \frac{2x}{1-x^2}$ with respect to $\sin^{-1} \frac{2x}{1+x^2}$. 4

3. a) If $y = \tan^{-1} x$ then show that, $(1+x^2)y_{n+1} + 2nxy_n + n^2y_{n-1} - ny_{n-1} = 0$ 6

- b) Verify Rolle's theorem for $f(x) = 2x^3 + x^2 - 4x - 2$ over $[-\sqrt{2}, \sqrt{2}]$ 4

Or

Test whether the mean value theorem is true for $f(x) = 3 + 2x - x^2$ on $(0, 1)$. 4



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Mid Term Examination

Program: B. Sc. in CSE
 Course Code: MATH-1107
 Time: 1:30 hours

Semester: Spring-2024
 Course Title: Mathematics-I
 Total Marks: 30

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks,
 (ii) Please answer the several parts of a question sequentially.
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Course Learning Outcomes (CLOs) of the Questions

CLO1:	Compute the functions, limit and continuity of a function, derivatives, integrals and extrema of single-variable and/or multivariable functions.
CLO2:	Understand the techniques of differentiation and integration.

Bloom's Taxonomy Domain Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

		Marks	CLOs	DLs
1. a)	Find the Maclaurin Series for the function $f(x) = \log(1+x)$	2	CLO1	U
b)	Find the domain and range of the function, $f(x) = x+3 + x-3 $	2	CLO1	U
c)	Test the differentiability of the following function at $x = \frac{3}{2}$	3	CLO1	App

$$f(x) = 3 + 2x; \quad -\frac{3}{2} \leq x < 0$$

$$= 3 - 2x; \quad 0 \leq x \leq \frac{3}{2}$$

$$= -3 - 2x; \quad x \geq \frac{3}{2}$$

Or

Test the differentiability of the following function at $x = 0$

$$f(x) = 1; \quad x < 0$$

$$= 1 + \sin x; \quad 0 \leq x < \frac{\pi}{2}$$

$$= 2 + \left(x - \frac{\pi}{2}\right)^2; \quad x \geq \frac{\pi}{2}$$

- d) Evaluate the limit using L'Hospitals rules for $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$ 3 CLO1 U
- Or
- Differentiate $\tan^{-1} \frac{2x}{1-x^2}$ with respect to $\sin^{-1} \frac{2x}{1+x^2}$.
2. a) Find the differential co-efficient of $\sin x$ by the first principle method. 3 CLO2 U
- b) If $y = \log \sec(ax+b)^3$, find $\frac{dy}{dx} = ?$ 3 CLO2 U
- c) The radius of a circle increases at a rate of 3 cm/sec. Find the rate of change of the area when i) the radius is 5 cm ii) the area is $4\pi \text{ cm}^2$ 4 CLO2 App
3. a) If $y = x^n \log x$; Prove that $y_{n+1} = \frac{n!}{x}$ 6 CLO2 App
- b) Suppose we are asked to determine whether Rolle 's Theorem can be applied to $f(x) = x^4 - 2x^2$ on the closed interval $[-2, 2]$. And if so, find all values of c in the interval that satisfies the theorem 4 CLO2 R&U
- Or
- Verify the Mean Value Theorem for $f(x) = 2x^3 - 8x + 1$ when $a = 1$ and $b = 3$ CLO2 R&U



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Semester: Autumn-2023
Course Title: Mathematics-I
Total Marks: 30

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
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Course Learning Outcomes (CLOs) of the Questions

CLO1:	Compute the functions, limit and continuity of a function, derivatives, integrals and extrema of single-variable and/or multivariable functions.
CLO2:	Understand the techniques of differentiation and integration.

Bloom's Taxonomy Domain Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Marks CLOs DLs

1. a) Obtain the Maclaurin's series generated by the function $f(x) = \cos x$
 b) Find the domain and range of the function,
 $f(x) = |x + 2| + |x - 2|$
 c) Test the differentiability of the following function at $x = 0$

2 CLO1 U
 2 CLO1 U
 3 CLO1 U

$$f(x) = 3 + 2x; \quad -\frac{3}{2} \leq x < 0$$

$$= 3 - 2x; \quad 0 \leq x \leq \frac{3}{2}$$

$$= -3 - 2x; \quad x \geq \frac{3}{2}$$

Or

Test the differentiability of the following function at $x = \frac{\pi}{2}$

$$f(x) = 1; \quad x < 0$$

$$= 1 + \sin x; \quad 0 \leq x < \frac{\pi}{2}$$

$$= 2 + \left(x - \frac{\pi}{2}\right)^2; \quad x \geq \frac{\pi}{2}$$

- d) Evaluate the limit using L'Hospitals rules $\lim_{x \rightarrow 0} \left(\frac{1}{x}\right)^{\tan x}$ 3 CLO1 U
 Or
 Differentiate $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ with respect to $\sec^{-1} \frac{1}{2x^2-1}$
2. a) Find the differential co-efficient of e^x by the first principle method. 3 CLO2 U
 b) Find the differential co-efficient of the following function $y = x^{\cos^{-1} x}$ 2 CLO2 U
 c) If $C = 1 + r \cos \theta + \frac{r^2 \cos 2\theta}{2!} + \frac{r^3 \cos 3\theta}{3!} + \dots$ 5 CLO2 U
 $S = r \sin \theta + \frac{r^2 \sin 2\theta}{2!} + \frac{r^3 \sin 3\theta}{3!} + \dots$
 Prove that $C \frac{dC}{dr} + S \frac{dS}{dr} = (C^2 + S^2) \cos \theta$
3. a) If $y = e^{a \sin^{-1} x}$ then prove that 6 CLO2 App
 i. $(1-x^2)y_2 - xy_1 - a^2 y = 0$
 ii. $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 + a^2)y_n = 0$
- b) State Rolle's theorem. Verify the Roll's theorem for the function 4 CLO2 R&U
 $f(x) = x^2 - 3x + 2$ in the interval (1,2)
 Or
 State Mean Value theorem. Verify Mean Value theorem for the CLO2 R&U
 function $f(x) = \frac{x^3}{4} + 1$ over the interval [0, 2].

International Islamic University Chittagong
 Department of Computer Science & Engineering
 Mid Term Examination (Spring Semester-2023)
 Mathematics I, MATH-1107

Marks-30

Time-1.30 hours

(Answer the following Questions)

1. a) Find the domain and range for the function $y = f(x) = \frac{x+1}{x-1}$ 2 CLO1
 Or
 Evaluate $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$ 2 CLO1
- b) Find the Maclaurin Polynomials for the function $f(x) = e^x$ 3 CLO1
 Or
 Test the continuity and differentiability for the functions at 3 CLO1
 $x = \frac{\pi}{2}$
 $f(x) = 1; \quad x < 0$
 If $f(x) = 1 + \sin x; \quad 0 \leq x < \frac{\pi}{2}$
 $f(x) = 2 + (x - \frac{\pi}{2})^2; \quad x \geq \frac{\pi}{2}$
- c) Suppose we are asked to determine whether Rolle's Theorem can be applied to $f(x) = x^4 - 2x^2$ on the closed interval $[-2, 2]$. And if so, find all values of c in the interval that satisfies the theorem. 5 CLO2
 Or
 Verify the Mean value theorem for the function 5 CLO2
 $f(x) = x^2 - 5x + 7$ in the interval $-1 \leq x \leq 3$
2. a) Differentiate $\tan^{-1} \frac{2x}{1-x^2}$ with respect to $\sin^{-1} \frac{2x}{1+x^2}$. 5 CLO2
 b) Differentiate $\cos x$ using first principle 5 CLO2
3. a) If $y = \sin^{-1} x$, then show that: 8 CLO2
 $y_{n+2}(1-x^2) - (2n+1)xy_{n+1} - (n^2)y_n = 0$
 b) Also find $(y_2)_0$ 2 CLO2



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B. Sc. in CSE, Mid Term Examination (Online), Spring-2022
Course Code: MATH-1107, Course Title: Mathematics-I

Time: 3:00 hours Marks: 21

(Answer all questions. Figures in the right margin indicates full marks)

Section: 1AM

[N.B. Please answer the several parts of a question sequentially]

SET: ALPHA

1. a) Test the continuity of the function, 4
 $f(x) = |x + a| + |x + (a + 1)|$ at the point $x = a$

Where a is the last digit of your ID

- b) Using L'Hospital's rules evaluate the limit: $\lim_{x \rightarrow \infty} x\{(a + 1)^{1/x} - 1\}$ 3

Where a is the last digit of your ID

2. a) Differentiate $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ with respect to $\sec^{-1} \frac{1}{2x^2-1}$ 4

- b) If $y = \frac{x}{(a+1)^+} \cdot \frac{x^2}{(a+1)^+} \cdot \frac{x}{(a+1)^+} \cdot \frac{x^2}{(a+1)^+} \cdot \dots \dots \dots \infty$ then find $\frac{dy}{dx}$ 3

Where a is the last digit of your ID

3. a) If $x = \sin\left(\frac{1}{(a+1)} \ln y\right)$ then using the Leibnitz theorem show that, 4
 $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - \{n^2 + (a + 1)^2\}y_n = 0$

Where a is the last digit of your ID

- b) Verify the truth of Mean Value theorem for the function, 3
 $f(x) = x^3 - 2x^2 + 3x + a$ in the interval $[0, (a + 1)]$

Where a is the last digit of your ID